

Q1 - 24 January - Shift 1

The maximum vertical height to which a man can throw a ball is 136 m. The maximum horizontal distance upto which he can throw the same ball is

- (1) 192 m (2) 136 m
(3) 272 m (4) 68 m

Space for your notes:

Q2 - 25 January - Shift 2

Two objects are projected with same velocity 'u' however at different angles α and β with the horizontal. If $\alpha + \beta = 90^\circ$, the ratio of horizontal range of the first object to the 2nd object will be :

- (1) 4 : 1 (2) 2 : 1
(3) 1 : 2 (4) 1 : 1

Space for your notes:

Q3 - 29 January - Shift 1

A stone is projected at angle 30° to the horizontal. The ratio of kinetic energy of the stone at point of projection to its kinetic energy at the highest point of flight will be :

- (1) 1 : 2 (2) 1 : 4
(3) 4 : 1 (4) 4 : 3

Space for your notes:

Q4 - 29 January - Shift 2

A particle of mass 100 g is projected at time $t = 0$ with a speed 20 ms^{-1} at an angle 45° to the horizontal as given in the figure. The magnitude of the angular momentum of the particle about the starting point at time $t = 2\text{s}$ is found to be

$\sqrt{K} \text{ kg m}^2 / \text{s}$. The value of K is _____.

(Take $g = 10 \text{ ms}^{-2}$)



Space for your notes:

Q5 - 31 January - Shift 1

The initial speed of a projectile fired from ground is u . At the highest point during its motion, the

speed of projectile is $\frac{\sqrt{3}}{2}u$. The time of flight of

the projectile is:

(1) $\frac{u}{2g}$

(2) $\frac{u}{g}$

(3) $\frac{2u}{g}$

(4) $\frac{\sqrt{3}u}{g}$

Space for your notes:

Q6 - 31 January - Shift 1

The speed of a swimmer is 4 km h^{-1} in still water.

If the swimmer makes his strokes normal to the flow of river of width 1 km , he reaches a point 750 m down the stream on the opposite bank.

The speed of the river water is _____ km h^{-1} .

Space for your notes:

Q7 - 31 January - Shift 2

Two bodies are projected from ground with same speeds 40 ms^{-1} at two different angles with respect to horizontal. The bodies were found to have same range. If one of the body was projected at an angle of 60° , with horizontal then sum of the maximum heights, attained by the two projectiles, is _____ m . (Given $g=10\text{ms}^{-2}$)

Space for your notes:

Q8 - 01 February - Shift 1

A child stands on the edge of the cliff 10 m above the ground and throws a stone horizontally with an initial speed of 5 ms^{-1} . Neglecting the air resistance, the speed with which the stone hits the ground will be _____ ms^{-1} (given, $g = 10 \text{ ms}^{-2}$).

(1) 20

(2) 15

(3) 30

(4) 25

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (3)

Q2 (4)

Q3 (4)

Q4 (800)

Q5 (2)

Q6 (3)

Q7 (80)

Q8 (2)

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Q1 (3)

$$H_{\max} = \frac{v^2}{2g} = 136\text{m}$$

$$R_{\max} = \frac{v^2}{g} = 2H_{\max}$$

$$= 2(136)$$
$$= 272 \text{ m}$$

Q2 (4)

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$$\text{Range} = \frac{u^2 \sin 2\theta}{g}$$

Range for projection angle “ α ”

$$R_1 = \frac{u^2 \sin 2\alpha}{g}$$

Range for projection angle “ β ”

$$R_2 = \frac{u^2 \sin 2\beta}{g}$$

$$\alpha + \beta = 90^\circ \text{ (Given)}$$

$$\Rightarrow \boxed{\beta = 90^\circ - \alpha}$$

$$R_2 = \frac{u^2 \sin 2(90^\circ - \alpha)}{g}$$

$$R_2 = \frac{u^2 \sin(180^\circ - 2\alpha)}{g}$$

$$R_2 = \frac{u^2 \sin 2\alpha}{g}$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{\left(\frac{u^2 \sin 2\alpha}{g} \right)}{\left(\frac{u^2 \sin 2\alpha}{g} \right)} = \frac{1}{1}$$

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Q3 (4)

$$\frac{KE_{\text{POP}}}{KE_{\text{top}}} = \frac{\frac{1}{2} M (u)^2}{\frac{1}{2} M (u \cos 30^\circ)^2} = \frac{4}{3}$$

Q4 (800)

$$\text{Use } \Delta L = \int_0^t \tau dt$$

$$L_0 = \int_0^2 mg(v_x t) dt$$

$$= mg v_x \frac{t^2}{2} = (0.1)(10)(10\sqrt{2}) \frac{2^2}{2}$$

$$= 20\sqrt{2}$$

$$= \sqrt{800} \text{ kg m}^2 / \text{s}$$

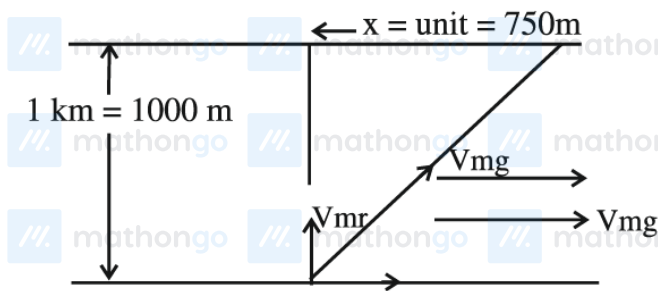
Q5 (2)

$$u \cos \theta = \frac{\sqrt{3}u}{2} \Rightarrow \cos \theta = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = 30^\circ$$

$$T = \frac{2u \sin 30^\circ}{g} = \frac{u}{g}$$

Q6 (3)



time to cross the River width $t = 1000 \text{ m}$

$$t = \frac{1 \text{ km}}{4 \text{ km/h}}$$

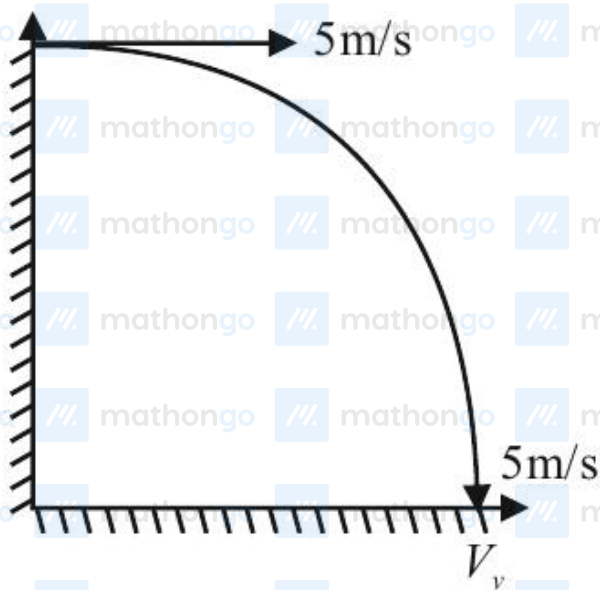
$$\text{Drift } x = V_{m/g} \times t$$

Where $V_{m/g}$ is velocity of River w.r. to ground.

Q7 (80)

No Solution

Q8 (2)



$$v_y = \sqrt{2gh} = \sqrt{200}$$

$$v_{net} = \sqrt{25 + 200} = 15 \text{ m/s}$$